lecture 19 Polynomial Interpolation: Lagrange Basis and Error Analysis @April 12, 2026

Summary

Executive Summary

- Polynomial interpolation constructs a polynomial that passes exactly through given data points, unlike least squares which minimizes error across points
- For $n+1$ distinct points, there exists a unique interpolating polynomial of degree at most n (existence and uniqueness theorem)
- Monomial basis ($1, x, x^2, ldots, x^n$) leads to ill-conditioned Vandermonde matrices and lacks meaningful parameter interpretation
- Lagrange basis polynomials $L_k(x)$ satisfy $L_k(x_i) = \delta_{ki}$ (1 at x_k, 0 elsewhere), enabling direct construction without solving linear systems
- Lagrange interpolation reduces construction cost from $O(n^3)$ to $O(n^2)$ and evaluation from $O(2n)$ to $O(5n)$ using barycentric weights
- Interpolation error for C^{n+1} functions is $|f(x) - p_n(x)| = \frac{|f^{(n+1)}(\psi(x))|}{(n+1)!} \prod_{i=0}^{n} |x - x_i|$ where $\psi(x) \in (a, b)$
- For equally spaced nodes with spacing h, error bound simplifies to $\frac{Mh^{n+1}}{4(n+1)}$ where M bounds $|f^{(n+1)}|$
- Equally spaced nodes are intuitive but not optimal (issue to be addressed in next lecture)

Conceptual Map

Polynomial Interpolation Problem Formulation

(A) Core Concept & Context

Polynomial interpolation constructs a polynomial $p_n(x)$ of degree at most n that satisfies $p_n(x_i) = f(x_i)$ for $n+1$ distinct data points (x_i, y_i). This is fundamentally different from least squares approximation: interpolation demands exact passage through data points (zero error at nodes), while least squares minimizes aggregate error when the system is overdetermined. Interpolation is appropriate when measurement error is small and the underlying function is believed to be smooth between sample points.

(B) Detailed Explanation

Given $n+1$ distinct points $\{(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)\}$, we seek $p_n(x) = \sum_{j=0}^{n} \theta_j \phi_j(x)$ where $\{\phi_j\}$ form a basis for polynomials of degree n. The interpolation conditions $p_n(x_i) = y_i$ yield a square $(n+1) \times (n+1)$ linear system. For distinct nodes, this system has full rank and a unique solution exists. The key choice is the basis: different bases yield the same polynomial (uniqueness) but differ dramatically in computational cost, numerical stability, and interpretability.

(C) Formal Notations (LaTeX)

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Key formulas

- Algebraic polynomial: $p_n(x) = \sum_{j=0}^{n} a_j x^j$ where $a_n \neq 0$
- Interpolation condition: $p_n(x_i) = y_i$ for $i = 0, 1, \ldots, n$
- Generic basis representation: $p_n(x) = \sum_{j=0}^{n} \theta_j \phi_j(x)$

Where: n = polynomial degree, x_i = distinct interpolation nodes, y_i = function values at nodes, θ_j = basis coefficients, $\phi_j(x)$ = basis functions.

(D) Critical Connections

- Extends earlier least squares work: same linear combination framework $\sum \theta_j \phi_j(x)$, but now the system is square and consistent rather than overdetermined
- Contrasts with Taylor polynomial approximation: Taylor series provide local accuracy near expansion point but lack uniform accuracy across an interval
- Relates to linear systems: interpolation requires solving $A\theta = y$ where matrix structure depends critically on basis choice

- Builds on Weierstrass approximation theorem: any continuous function on $[a, b]$ can be uniformly approximated by polynomials, justifying the approach

(E) Evidence from the transcript

"when you Use least squares. You don't want necessarily your fitting function to go through the data points... where is it when you're doing interpolation, this is something that you have to have because this is the definition of interpolation"

"if you want a polynomial interpolation in a neighborhood of just one point, I will go with Taylor polynomial. But if you were back-defeating distinct points... you want a uniform good approximation in a big interval"

"the polynomial, the interpolant polynomial exists and is unique... it doesn't really matter how you get there"

Monomial Basis Approach and Its Limitations

(A) Core Concept & Context

The monomial basis $\{1, x, x^2, \ldots, x^n\}$ is the most obvious choice for constructing polynomials, leading to the natural representation $p_n(x) = theta_0 + theta_1 x + ldots + theta_n x^n$. However, this basis produces a Vandermonde matrix A with entries $A_{ij} = x_i^j$, which is notoriously ill-conditioned for large n or widely spaced nodes. The resulting system is solvable but numerically unstable, and the coefficients θ_j lack physical interpretation.

(B) Detailed Explanation

For the monomial basis, the interpolation matrix has the structure where row i contains $[1, x_i, x_i^2, ldots, x_i^n]$. Solving $A\theta = y$ costs $O(n^3/3)$ operations via Gaussian elimination. Polynomial evaluation using Horner's nested scheme costs $O(2n)$ flops. Two critical flaws emerge: (1) The parameters θ_j have no direct relationship to the data values y_i — changing one data point propagates through all coefficients, making the problem ill-conditioned to data perturbations. (2) The Vandermonde matrix has high condition number, especially for large n or spread-out nodes, leading to numerical instability.

(C) Formal Notations (LaTeX)

∑ **Key formulas**

- Vandermonde matrix: $A = \begin{bmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{bmatrix}$
- System to solve: $A\theta = y$ where $theta = [theta_0, theta_1, ldots, theta_n]^T$, $y = [y_0, y_1, \ldots, y_n]^T$

Where: A = Vandermonde matrix, θ_j = monomial basis coefficients (not equal to y_j).

(D) Critical Connections

- Same linear system structure as least squares but square instead of overdetermined
- Requires linear system solution methods (Gaussian elimination, LU decomposition) previously studied
- Polynomial evaluation connects to Horner's algorithm for efficient nested computation
- Matrix condition number concepts apply: high condition number of Vandermonde matrix causes numerical difficulties

(E) Evidence from the transcript

"A is a Vandermonde matrix... this is a matrix that has a very high, usually a very high condition number, and people don't like it"

"the parameters that you found are not related to the value of the function at the point. So my theta_i is not related to the value of the function at x_i"

"You change one of the data points very slightly, you're going to change the whole polynomial. So your problem is actually ill-conditioned"

Lagrange Interpolating Polynomials

(A) Core Concept & Context

Lagrange interpolating polynomials exploit a brilliant basis choice: for each node x_k, construct a polynomial $L_k(x)$ that equals 1 at x_k and 0 at all other nodes. This cardinal property $L_k(x_i) = \delta_{ki}$ transforms the interpolation

problem from solving a linear system to simple weighted summation: $p_n(x) = sum_{k=0}^{n} y_k L_k(x)$. The coefficients are exactly the data values, making the method numerically stable and highly interpretable.

(B) Detailed Explanation

The Lagrange basis polynomial for node x_k has the form $L_k(x) = prod_{j=0, j neq k}^{n} frac{x - x_j}{x_k - x_j}$. The numerator vanishes at all nodes except x_k, while the denominator normalizes to give $L_k(x_k) = 1$. Each L_k is a degree-n polynomial with n zeros (at all nodes except x_k) and $n - 1$ extrema alternating between minima and maxima. The value 1 attained at x_k is not necessarily the global maximum.

For efficient computation, use barycentric form: precompute weights $w_k = \prod_{j \neq k} \frac{1}{x_k - x_j}$ once (cost $O(n^2)$), then evaluate $p_n(x) = frac{sum_{k=0}^{n} frac{w_k y_k}{x - x_k}}{sum_{k=0}^{n} frac{w_k}{x - x_k}}$. This reduces evaluation to $O(5n)$ operations. Special care needed: if evaluating at a node $x = x_k$, return y_k directly to avoid division by zero.

(C) Formal Notations (LaTeX)

> ∑ **Key formulas**
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> - Lagrange basis polynomial: $L_k(x) = \prod_{j=0, j \neq k}^{n} \frac{x - x_j}{x_k - x_j}$
> - Cardinal property: $L_k(x_i) = \begin{cases} 1 & \text{if } i = k \\ 0 & \text{if } i \neq k \end{cases}$
> - Interpolant: $p_n(x) = \sum_{k=0}^{n} y_k L_k(x)$
> - Barycentric weights: $w_k = \prod_{j=0, j \neq k}^{n} \frac{1}{x_k - x_j}$

Where: $L_k(x)$ = Lagrange basis polynomial for node k, w_k = barycentric weight (constant for given node set), y_k = data value at node k.

(D) Critical Connections

- Eliminates need for solving linear systems — the interpolation matrix becomes the identity
- Coefficients now directly equal data values y_k, unlike monomial basis where θ_j are opaque

- Same uniqueness guarantee as monomial basis: different bases yield algebraically identical polynomials
- Numerical stability vastly improved: no Vandermonde matrix, no ill-conditioning from matrix inversion

(E) Evidence from the transcript

"And why is this good? Because remember that the property of the polynomial... it has to go through the points, through the data points... So, Once you find such a polynomial, the previous theorem says that's the right polynomial because it has the right Degree and it has the property that it goes through the points"

"I think this is such a brilliant idea. And I don't have to solve any system of equations"

"the matrix of the system that we need to solve is just the identity. Because... all the These guys here in the diagonal are one, and every time you calculate the value of the Lagrange polynomial at the distinct point, you're going to get a zero"

Interpolation Error Analysis

(A) Core Concept & Context

While the interpolant exactly matches the function at nodes ($f(x_i) = p_n(x_i)$), error exists between nodes. Quantifying this error is essential for assessing approximation quality and choosing polynomial degree. Two theorems bound the error: a general result (Theorem 1) valid for any distinct nodes, and a sharper bound (Theorem 2) for equally spaced nodes. Both require the function f to have $n+1$ continuous derivatives.

(B) Detailed Explanation

The first interpolation error theorem, proven using Rolle's theorem, expresses error as $f(x) - p_n(x) = \frac{f^{(n+1)}(\psi(x))}{(n+1)!} \prod_{i=0}^{n}(x - x_i)$ where $\psi(x) \in (a, b)$ depends on x but is unknown. This provides an exact representation but is not directly computable since $\psi(x)$ is unknowable. However, if we bound $|f^{(n+1)}(\xi)| \leq M$ for all $xi in [a,b]$, we obtain $|f(x) - p_n(x)| leq frac{M}{(n+1)!} prod_{i=0}^{n}|x - x_i|$.

For equally spaced nodes with spacing $h = frac{b-a}{n}$, a technical result bounds $prod_{i=0}^{n}|x - x_i| leq frac{h^{n+1} n!}{4}$, yielding the sharper bound $|f(x) - p_n(x)| leq frac{Mh^{n+1}}{4(n+1)}$. This shows error decreases

with smaller spacing and is bounded uniformly across the interval. However, equally spaced nodes are not optimal (to be discussed in next lecture).

(C) Formal Notations (LaTeX)

> ∑ **Key formulas**
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> - First interpolation error theorem: $f(x) = p_n(x) + \frac{f^{(n+1)}(\psi(x))}{(n+1)!} \prod_{i=0}^{n}(x - x_i)$
> - Error bound (general): $|f(x) - p_n(x)| \leq \frac{M}{(n+1)!} \prod_{i=0}^{n} |x - x_i|$
> - Second theorem (equally spaced): $|f(x) - p_n(x)| \leq \frac{Mh^{n+1}}{4(n+1)}$

Where: $\psi(x) \in (a, b)$ = unknown point depending on x, $M = max_{xi in [a,b]} |f^{(n+1)}(xi)|$, $h = \frac{b-a}{n}$ = node spacing.

(D) Critical Connections

- Relies on Rolle's theorem from calculus (existence of intermediate points where derivatives vanish)
- Error formula resembles Taylor remainder term but applies globally across the interval
- Requires function smoothness (C^{n+1}) — more derivatives available enables higher-order accurate interpolation
- Spacing h appears with exponent $n+1$: doubling nodes (halving h) reduces error by factor 2^{n+1}

(E) Evidence from the transcript

"what is the error in replacing sine of x by its interpolating polynomial... constructed from 10 equally spaced nodes in that particular strange interval"

"P9 approximates the sine with at least eight decimal digits of accuracy"

"next time... I will explain why we don't use equally spaced" (nodes — implying limitations of uniform spacing)

Comparisons and supporting details

Aspect	Monomial Basis
Basis functions	$1, x, x^2, \ldots, x^n$
Coefficient meaning	Opaque θ_j unrelated to data
System matrix	
Construction cost	$O(n^3/3)$ (solve linear system)
Evaluation cost	$O(2n)$ (Horner's method)
Numerical stability	Poor for large n
Data sensitivity	Changing one point affects
Implementation	Requires linear solver
Comparison	Polynomial Interpolation
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Goal	Exact passage through all data points
System	Square $(n+1) \times (n+1)$
Solution	Unique solution exists (for distinct nodes)
Error at nodes	Zero by definition
When to use	Small measurement error, smooth underlying function
Degrees of freedom	Number of points =
Comparison	Taylor Polynomial
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Accuracy region	Local (near expansion point x_0)
Basis	Powers $(x - x_0)^k$
Information used	Derivatives at single point
Error growth	Increases rapidly away from x_0
Uniform approximation	No

Action Items

☐ Next lecture will address optimal node placement (non-equally-spaced nodes)

Notes

Transcript

that I had before. So the function, the equations that I'm solving in newtons usually is F equals zero. So f equals 0. But now my f-is the gradient of phi because I'm trying to find the critical points. So what I'm solving is in Newton's terms, what I'm solving is F equals to zero F is nonlinear in the parameters. But what I'm actually doing in terms of least squares is to minimize the phi. So I need to find the critical values or the critical points for the phi.

So I need to find, to solve the gradient that phi is equal to zero.

If we Use the nomenclature that we found Last night So last time I had this equation here, and this was the equation for the iteration of newtons. So this guy here is the Jacobian. Calculated iteration k, that's going to be applied to x of k plus 1 minus x of k.

This is my delta x, so that I know which step I'm going to go from the current iterate to the next iterate. equals to the negative of f. But now my f is actually the gradient of phi. So if you rewrite newtons in terms of phi instead of the generalized f, Every time you see an F, you should have a gradient of phi. So when you differentiate the function, you are differentiating the gradient of phi, so you're getting the S here, phi. Okay.

So the S-yant, the matrix of the second-order partial derivatives, at that particular iterate. So this here is this matrix here when your F is the gradient of phi. And on the right-hand side, because your F is the gradient of T, you just have minus the gradient of U. So you just need to solve this.

Well, Jess... This is not a big deal, but it has its issues.

So the first thing is that you have to assureThat, um... the Ayr-Zehl matrix actually exists.

And after you are sure that it exists, you have to be able to calculate it.

So this is one of the issues they can usually have.

And your STN is not necessarily positive definite if you are far from the minimizer of pi that you're searching for. So that might also become a problem.

But anyway, I am solving a system of linear equations at each one of the iterations.

So this is something that--It's good.

Okay, so As I mentioned, this is not--Heart?

in abstraction, but you need to put together a lot of things that we've been learning in the past few weeks to kind of put them together to solve one

particular problem.

So if you are a bit amused about this, it's because you might not be up to date with what we have been doing in the past few lectures.

But I'm happy to take any questions regarding. So this is the wrapping up. nonlinear equations and in this case using Newton's method to solve nonlinear least squares. So it's a data feeding problem. So on the Still on that, the fitting, I'm going to change gears to something completely different. So we have been looking at these squares lately. And we needed nonlinear equations to solve nonlinear least squares.

But actually, when we were looking into these squares, if you guys remember, I think my first slide about these squares was about is approximating a function by least squares. different from approximating a function by polynomial interpolation. And at the time I said that the big difference was that when you Use least squares. You don't... You don't want necessarily your fitting function to go through the data points.

Okay, where is it when you're doing interpolation, this is something that you have to have because this is the definition of interpolation. So I am going to start today's lecture or after this wrapping up. with the polynomial.

Okay, so this is my goal.

I have a set of distinct points. And the idea there of distinct is relevant. because you're going to see that a lot of the theories and the existence and uniqueness and error theorems that we have regarding polynomial perturbation, they actually require you to have distinct points, otherwise everything goes up. So I'm assuming that I have distinct points. So the Xi's that you see there, Well, the f are distinct points.

And what I want to know, what I want to do is I want to come up with a polynomial function that goes through all these points. And, um... So the interpolation points, the error has to be zero. The error between the data and the value of the polynomial that we put. So this is what we have been doing. I have here the definition of an algebraic polynomial. This is a function that has that form. Notice that I'm going to go from...

The 0th order term, so 1, so x to the power 0, multiplied by a0. all the way to this a_x_n. Um... where your M is a positive integer, Your A's So your coefficients here are real constants. And because the highest order monomial assuming that my a is different from zero is x of n, we say that this is a

polynomial of order n. Or a polynomial of n order. Okay, so the question is, how do we construct the polynomial approximation?

What are reasonable ways of doing it? What is the error in their approximation? And is there a better polynomial interpolation than another. So all these questions we need to have answered. But before I go on, I just want to make sure that everybody understands the difference between getting a polynomial least squares and being polynomial interpolation. So they are completely different because in the polynomial interpolation, what I need to do is to come up with a function that goes through the points.

And this is usually done if you are pretty sure that the error of your data points is small, Where is you going to use least squares if you are not sure that the data has large or small pairs. Okay? Because whatever you do, you're going to go, we did triple our ink, but... The logic is you don't want to go through a point if the point is known to have a lot of error. So you're basically looking not at the point itself, but at a pattern, whereas here we are just looking at the point itself. So this is Uh... The topic of today or of this next these next few lectures. Okay, so why polynomials?

I don't know if you, so we like polynomials.

Basically because we know well polynomials and when I say know well Um... There's a theorem or a result which is called the Weizentraust approximation theorem that tells that a polynomial can approximate any continuous function uniformly. So this means that the error is basically uniform or it has a uniform way of approaching the real continuous function. So what I mean by this is when I say uniform, I mean When I say uniformly, This is the adverb that is contrary to or in opposition with the point wise.

Because you can have a pointwise approximation, so this is basically saying that everything is very accurate at certain points, but You know, outside those points, you don't really know what's going on. So this is a pointwise approximation. But on the other hand, if you want a uniform approximation, you're saying that the error is bounded across all the points that you are considering. So this is one of the reasons why we like polynomial interpolation because this guy, Bars and Streis, I think there's too many S's in that name and too many E's.

I don't know what happened, but it's usually shorter. I think it doesn't have, I don't think it has... This is S and this is E. But now... The thing is I don't

memorize words. I don't find, I memorize words by what they look like. I don't know if you always do this. I don't think people do this.

So if I find a new word, I really have to read it. Otherwise, I just guess what it is. And this is what I've done. I've written down guessing and probably didn't check it out. It seemed more or less OK. It seems right. It seems right.

I mean, I checked. Yeah, it's right. Yeah.

But the point is, this guy said that if I have a continuous function, then I can uniformly approximate that function with a polynomial. Okay, so this is one big plus advantage if you're gonna go into a polynomial approximation.

The other big plus is that you know how to integrate well polynomials and you know how to differentiate well polynomials. So if you have a continuous function and you want to differentiate or approximate it, maybe the first thing that you differentiate or integrate it, so maybe the first thing that you can do is if you don't know how to do that, you can just look at its polynomial, interpolant polynomial.

Okay, so the other possibility would be instead of the instead of approximating a function, a continuous function by a polynomial, you can still see, look and say, oh look, what I'm gonna do is I'm just gonna tailor expand this function And I have a polynomial just by Taylor expanded, right? But if you guys remember, the Taylor expansion is only valid and accurate in a nearby, in a very close neighborhood to a certain point.

So you would not be having, if you're Taylor expand about one point, so if you're doing a A polynomial approximation of a function about one point, you don't see what's going on. in a part of the interval that it's far from that particular point. So you would not have a uniformly good approximation in the interval of interest. Okay? So this is why Theta polynomials are not used in interpolation because they require something that is locally Good, but it's not globally good as in the case of interpolating polynomials.

Okay, so this is the picture.

Okay, so if you want to approximate e to the power x, and you want to approximate it about x equals 0, you can see that you have here, so this is the function itself, the exponential. This is the first order approximation. This is the Taylor. First order, well, 0 order, first order, second order, third order, and so on. And you can see that the higher the order of the Taylor polynomial, you're getting further and further away from the value of the function.

So the function Yes.

This guy here, and it approximates very well in this particular point, but it doesn't do so when you go further and further away from the function. from the point where you are They were expanding, which in this case is X naught.

Okay, so...

This is why we don't use, I mean, Don't get me wrong. If you want a polynomial interpolation in a neighborhood of just one point, I will go with Taylor polynomial. But if you were back-defeating distinct points, as I remember, you want a uniform good approximation in a big interval. So you don't want to go with Taylor polynomial because you would lose the global uniformity, the uniformly good approximation of perception.

So, Okay, so when would I need a polynomial interpolant? So the polynomial that interpolates, it's called the interpolator or the polynomial interpolant or the interpolant. The idea of interpolate, so the word interpolate means that you go through the points. This is, I mean, I've set down here three possible explanations, so why we want to approximate data points with a function. is because many times you have discrete data points, but you want to Do operations on that data in a way that is a continuous set of points which you don't have.

So you would like to have a continuous function that you don't have, so you have to interpolate it. So if you want to plot, differentiate, or integrate a certain amount of data, you first need to interpolate the data. And another Another use for the interpolant is to actually determine with a certain degree of accuracy the values of the function that you are trying to interpolate from the data at different points other than the data points.

So many times you want to know the values of the system or the output of the system in between two inputs. So you need to interpolate and then calculate the value of the function at the point. Okay, so. We have, when we're doing interpolation, we usually have a wish list. Of all the good things that would like to happen So for example, you want the reasonable, so the P of n there, so you want the reasonable polynomial. This is P of n is the polynomial of order n.

So you want the reasonable polynomial. So you have here on the left and on the right, you have two. Raps? And they have the same bathroom. And then on the left, you have a little order polynomial and And on the right hand side, you have a polynomial with a higher order because you have a higher number of

oscillations. And then the question is, Um... Do you prefer the interpulent on the left? or the interpulent on the right.

So this is one question that's very reasonable to ask.

Um... So this is something that I would like to see answered for me. I would say that I prefer the one on the left because it doesn't imagine what is happening between for example point number two and point number three right it doesn't it doesn't imagine Whereas this guy here goes very further away between point one and two, and then it says that function oscillates a lot between two and three and so on. So this is maybe far-fetching of what's going on.

So I would go with the one on the left. On the other hand, your P of n should be pretty inexpensive to construct. Firstly, and also to evaluate, okay, but we're looking into the construction mainly today. The evaluation is also important. then the construction itself should be Numerically stable, but I would add also it should be meaningful to whatever we're doing, and you'll see what I mean with meaningful.

So before I go on, have you ever worked with interpulants? In your academic life, I guess? In different ways. In different ways? It's always different.

What do you mean? For example, in thermodynamics, we just do simple linear interpolation between two data points and define the middle point between them. Some of the courses we fit a polynomial and then use that equation to evaluate the input.

Yes, so this is one of the things that I was saying. You may need the value of the original function if a certain content is not in the data set. All the courses will do the same thing as the oneto make the fake pass through all of the points for error and then you have it.

The point that the that we need. So it's always different. Oh, you mean always different applications? Yes, always different applications and always different approaches. Oh, all these different coaches. Oh, good, good, good. Okay. Because there are smart approaches. lots of smart approaches, and hopefully I can tell-Sorry? That's why we, yeah.

Let's see. I think it's pretty obvious what is smart and what's not smart.

But I will spend the whole hour that I have trying to convince you that there is...

A smart approach, let's say. Well, it's not my approach, but you can imagine. Okay. Okay, so, Those are those this was my wish list. I think it's a pretty reasonable wish list, but if you're a mathematician You want to ask a little bit

more, you know how these things go because you don't want to have to intervalent polynomials of the same order.

So my first question, if I have to put my hat is, is my polynomial unique given Given a certain set of n plus 1 points, this is true. And we will have a glimpse of why this is true. I will not produce the proof, but I will show you that there is a reasonable reason, there's a reasonable reason to expect this to be true. Is there always a, so the first question is it unique? So if you give me... three points and you ask me for a polynomial of, so you give me three points, so n plus one is equal to three.

And you ask me, is this second order polynomial, interpulent polynomial, so is this polynomial that passes through these three points unique? Yes. And it doesn't matter How the smartest strategy you have to actually get to it, but there are smarter strategies in the sense of they make more sense, You can see actually what's happening. and they take less, they are less costly to actually construct in terms of getting to the.

Is there always an interpulent? Yes, there is always an interpulent. So the answer is yes, the interpulent is unique and it exists. And then how many times can a degree n polynomial oscillate? How do we choose? So how do we choose the number or the degree of the polynomial that I'm going to come up with? If my end is not large enough, then my quality of approximation may be poor, but if it's If n is too large, then I'm going to get too many oscillations.

And can I come up, and this is a big, big thing in polynomial interpolation, can I come up with a way to choose, if I can, of course, to choose the data points so that the magnitude of oscillations is the decrease Okay, so this is all things that we're gonna take a look at. Okay, so the formula is very easy to understand, I guess. So I'm going to consider-so remember, I have polynomials. So the easiest thing to do is to consider a linear combination of functions that form a basis have the polynomials of that order. So I want to know my function f.

So this is what I really want to know. But what I'm going to do is I'm going to come up with a polynomial in turbulence. So this is my p of n. that happens to be a linear combination of polynomials. And the linear combination that you see there has coefficients theta.

So we've seen this before, right?

But this is a completely different-It might look like we're doing least squares, but we are not. We are just taking the same modeling approach in the beginning. And then I want to come up so I know I am giving this data so I know

the values of my function at all the distinct points. Okay, so if I plug them into the function that you have there on the right hand side, your unknowns are the Thetas. And you want this to be equal to the values of the data points.

So this is what you have. So this is, you are given the Ys if you remember from our notation.

So you are given the y's at the axis. Okay. And you want that-The This equation that you have here, when you plug in the x's, you're going to get the 1's. So if you go to that function there, so the sum, and you say that my x is going to be equal to x naught, then you're going to have this first equation. So this sum is going to be equal to the value of the The data that you get that you're calling y0 at the point x0.

Yes.

I'm confused. Is this just coming on in V square fitting into a different hat?

No, so it's the same fitting, but you ask different things. So if you're doing If you're doing least squares, If you're doing these squares, the number of points that you have If you are given, has nothing to do in principle with the basis that you get. Here you are asking for this to beExactly. So when you are doing these squares, you want the sum of the squares of the errors to minimize. You want the graph of the function that you're searching for, so the graph of the polynomial, to go exactly through the points.

So this is to, one is the optimization problem, you want the minimization.

The other one is to solve this system.

So we just want to see the trend? No, no, no, no, the trend. You don't want the trend. You want bigs. You want...

I'm not sure what you mean with the trend, but here is more than trend. You want your polynomial, the polynomial that you're trying to find to go exactly through the points that you are given. Whereas in least squares, you don't require this to be true.

In fact, you know that this system of equations, which is the same, does not have a solution.

So instead of...

Solving this system that does not have a solution.

What you do is you project the right hand side into the column space of the matrix A.

Whereas here you require for the existence, well, existence is uniqueness of solution and this is why the polynomial existence is unique.

Okay, so there are two different approaches.

I know it may look the same, until here is the same. All right?

Until here is the same. Now, if I were to do these squares, So a couple of weeks ago, I would say this system has no solution.

So instead of trying to solve this guy that doesn't have a solution, what I'm going to do is I'm going to look for something similar.

And then you think either calculus or linear algebra. If you're thinking calculus, you're going to ask, For the...

The squares of the errors Do we minimize?

So this is calculus. If you're doing linear algebra, you're gonna ask for this guy. project like an orthogonal projection into the column space of this matrix A and then you have a solution but you are no longer solving the system for equal to this, you're solving the system for something that is very close to this. This is least squares. But in this case, you don't require-so you can have, for example, three points and you're fitting these three points to a line. That's not possible. I mean, in general, right? Unless they are already in a line.

And then if they are already on the line, then the system is just-it's true. It exists. It has a solution.

Okay, so now you're going to see, so this is least squares. If I'm doing linear interpolation, my thought process is completely different.

What I'm going to do is I'm going to say that there's a minimum number of points that I'm going to require For a polynomial of a given order. Okay, and in this case you will see that if you look at what's happening here, you see that you have The same number of points, I mean you have one plus how many points? N plus one point. Okay. And you have n plus 1 equations. So you have One equation, sorry, n plus one equations.

And you have n, n plus 1, n-nulls.

So this is a square matrix. So there's a lot of chance that this matrix has a solution.

And of course, if this matrix is n plus 1 by n plus 1, Okay, what happens is Remember the points are distinct.

So there's a big chance that these lines or columns are linearly independent. The matrix has an inverse. So the system has a solution. Because the inverse exists And the solution has to be unique. So this is why the polynomial, the intervalence, exists and is unique.

Is this okay? Guys, did you understand the difference between the least squares and the polynomial interpolation? Okay.

So, example. So this is the stupid example, but it's a good example because we understand why the other example is smart.

Let's say I have n equals to 3. Okay, and I have... My basis functions, so remember if I'm doing a polynomial interpolation and I don't know better, I'm going to ask for my basis function to be 1x x squared next cube, because this is the obvious solution. And then I have those points, 2, 14, 6, 24, 4, 35, and 7, 15. I build my A. So what is in this A? It's the value of the F, so the basis function at each one of the points.

If you go back to the previous slide,This eye here. It's gonna be... The value of the function, in my case, was 1.

Evaluate it at each one of the points. But because it's always 1, the first column is going to be 1.

Okay, so this is going to be What?

And then here, this is going to be x. So it's going to be 2, 6, 4, and 7. and then x squared and then x cubed. Okay, and my y, so my right hand side is gonna be 14, so the second entry in the data points.

And then, what will I just ask? For theta, and I get these numbers.

So I've solved the system a theta equals to the y, and I get these numbers. And then I just construct a polynomial. The interpolant is P of 3 because I have 4 points.

So if I want to have a system that is 4 by 4, I'm going to have an interpolant that has order 3 because you are always counting with the 0, x to the power 0, so with 1. And then I just plug, so this is going to be, this theta zero is going to be this number, theta one is 1.7, theta two is this guy, and then theta three is this guy.

So this is very easy to understand, I think. Okay. But then, It's very easy to implement. So it's very easy. You just need the solver AX equals B. However, the problem is, one of the problems, I don't think this would be like the biggest problem, but it's a problem, is that A is a Vandermonde matrix.

So it's the matrix that goes, well, it is just what I said.

It's 1x, x^2, x^3, x^4 evaluated at different That's the points.

And this is a matrix that has a very high, usually a very high condition number, and people don't like it.

You can calculate the number of operations that you need to construct the polynomial. And you can see that of course it's O and cubed over 3.

because you just need to solve the linear system of equations.

And then every time you want to evaluate the polynomial, You need O of 2n flops. If you do nested So this here is from solving a system of equations.

We learned this because it's a linear system of equations. So we know the, like the computational cost.

And we also saw how to come up with evaluating polynomials at a certain point using Horner's algorithm. So nested implementation.

And this was O to M. Okay, so this is still not a big problem from my point of view, especially now that everything runs very fast.

There are bigger problems. For me, one of the problems is that you can't, this is maybe not obvious from what I'm going to say now, but you will see when I come up with the solution, you will see exactly what I mean.

is that the parameters that you found are not related to the value of the function at the point.

So my theta_i is not related to the value of the function at x_i.

And you will again see what I mean exactly.

But for now just let it stay in your brains. And the other thing that we don't, so we don't like this, And what we don't like is that for some reason, You change one of the data points very slightly, you're going to change the whole polynomial.

So your problem is actually ill-conditioned because a small perturbation in the data set It's going to be propagated and we'll see that you will have a completely different set of fingers.

So we don't like this too.

And this is why the monomial basis is not a good basis because of, well, mainly I can just say because A is a Van der Waal matrix that has a lot of problems. But my conclusion is we need a better basis and a guy that's always in our lives,

Lagrange, he came up with a good thing. But before I... I will wait. To all do this, but I'm going to say it again. So now we're solving-so let me go back. So this is the theorem that-Again, not gonna show it. This is chapter 10 of "Asher and Grief," if I'm not mistaken.

There's a theorem that guarantees the existence and uniqueness of the interpolating polynomial. And the theorem says if you are given any real data points, Again?

With the xi is distinct, so they are different.

So you really have n plus 1 dot the points. Then there's the unique polynomial p. have degree at most n such that it is an interpolating polynomial. So the value of the polynomial at the xi is equal to the value of the function or the data points. Thank you. And why this is true?

Well, I can go back and tell you what I told you guys before, is that you have n plus 1 by n plus 1 matrix. And the rank of the matrix is m plus 1 because the points are distinct, so the values of the function f, of the basis function at the f are distinct. Okay, remember that the functions they have to be Because they are basis functions, they have to be linearly dependent.

So you're going to have rank n plus 1. The determinant is different from 0. The inverse of a exists.

So the solution to this problem, which are the things that we are searching for, exists and is unique.

So this is basically like the linear, I don't remember how Asher and Reif,You win? If they take the linear algebra route or if they take the calculus route, I think this is a personal preference. But the point is The polynomial, the interpolant polynomial exists and is unique.

And because it is unique, it doesn't really matter how you get there. So... Remember this polynomial that I got here?

If you have the same data points, But you choose a different basis, whatever basis you choose. I don't know, 1, x minus 1. I mean, there are different ways of choosing the basis. You will always get the same polynomial. So if you rearrange The polynomial, I mean in terms of algebraic calculations, you always get the same polynomial.

So what is the smart way? And I think this is one I think I'm always in awe every time I come back to this because I think this is such a smart idea. So I'm going to tell you what is the idea.

of Lagrange interpolating polynomials. I'm going to do it for two points, and then I will upgrade it to n plus 1 points. So if you are given two points, x naught, y naught, x1 and y1. I want to come up. with the first order polynomial at these two points, right?

And I'm going to define two functions that I'm going to call L0. And the other one.

That look, you know, look. Very easy. They are linear functions, x minus x1, x0 minus x1, and then for L1, x minus x0, x1 minus x0. 0 or x naught and the properties of these Lagrange polynomials is that if you look at what happens when you calculate L naught and x naught, you're going to get a 1. If you calculate the L0 at x1, you're going to have 0. And likewise, if you Calculate L1 at the other point, which is x0. You're going to have 0.

But if you calculate L1 and X1, you're going to get a 1. So these lag-range polynomials They are...

They are specific to one particular point.

So you have one of these guys per point that you are given in your data set. And at that particular point, there are one Always. And they are zero. at all the other points.

In the day to say. So... This is amazing. Why is this amazing?

Because then you don't even have to calculate. If you want to calculate the polynomial, you don't even have... to solve any system of linear equations. Because you just Your interpolating polynomial is just the value of the function at 0, so at x0, is going to be multiplied by L0. plus y1, l1, x. And why is this good? Because remember that the property of the polynomial, so it's a first-order polynomial, And it has to go through the points, through the data points, right?

So, Once you find such a polynomial, the previous theorem says that's the right polynomial because it has the right Degree and it has the property that it goes through the points. So what I mean is You are given the data. There exists a unique polynomial of a certain order.

such that it is an interpolating polynomial.

And because it exists and is unique, if I find one, it's the one that I was looking for. And Lagrange said, "Okay." So, if My L functions, those are the basis functions that I will be using to construct my polynomials. Have this properties I'm going to use these properties to come up with a polynomial without having

to solve a system of equations. So if you take the polynomial at x naught, This is going to be zero.

This is going to be one, and I'm getting the value that I want, which is why not? It goes through that particular point.

Likewise, so in this case, I only have two points, right?

Likewise, if I calculate the value of p of the polynomial x1, I should have y1.

Do I have it? Yes. Because this is going to be 0, and this is going to be 1.

So I have found the pulling oil.

I think this is such a brilliant idea.

And I don't have to solve any system of equations.

So the hard part is actually to show that the theorem is true. Because there's a little bit of, I've told you the idea, right? But there's a little bit of... the technical nuances that you have to do. But once you find it, it's unique. You can actually guess it. And that's the right one.

Okay, so this was for 2.8. Do you understand the idea? Do you understand how amazing this idea is?

I think it's really cool. Okay, so I'm going to do the same thing, so it's the same idea.

If you are given n plus 1 points, You're going to have a polynomial L of n, so of order n. corresponding to the point K that looks like what you see there. Okay.

And it has the same properties that we had before.

So if you evaluated at xi when the i is different from this index k, You're going to have a 1.

Otherwise, if your i is different from the k, you're going to have a 0. So it's the same properties that we have before.

And then if I want to construct the polynomial, the only thing I need to do is I need to multiply each one of the Lagrange interpolating polynomials, which are the L's. by the corresponding value of the data point of the data set that you have at that particular x_i.

So this slide that you see here It's just like an upgraded version for n plus 1 points of the slide that I had before. If you understand this, Going to M plus one point six. Okay, so moving on.

So these are the properties. I have the light branch interpolating polynomials. So the first one is that L is-so the wide range interpolating polynomial is 1.

Uh...

when you are at the polynomial that corresponds to the point that you are considering and zero all the other points It can be shown that that polynomial, so L_k, has n zeros. Well, because it's a polynomial of n order, so it has n zeros. This is the fundamental theorem of calculus. And it has n minus 1 extrema, and it alternates between minima and maxima. It is also true that the value 1, that the polynomial attains at the corresponding point, it doesn't have to be the maximum, so the polynomial can have Higher values.

But you know that at this particular point, at this particular xk, it's going to be 1.

I will show you the picture because I just listed here. And now the matrix of the system that we need to solve is just the identity. Because Remember that the system that I needed to solve Was this?

Okay, now my Fs, these Fs are the Lagrange interpolating polynomials. And they are only going to be one when you are at the corresponding x-axis. So all the These guys here in the diagonal are one, and every time you calculate the value of the Lagrange polynomial at the distinct point, you're going to get a zero.

So this is why you don't have to solve any system. Well, this is why the coefficients are exactly the values of the function at the points. So an example, I don't know if you have ever seen these guys.

So for this particular case of The four points that we had in the previous example You have four Lagrange interpolating polynomials because you have four points.

And you can guess which one. This, so the blue one, you can see that the blue one here, the blue graph, Is zero at seven? It's zero at six.

It's 0 at 4, but it's not 0 at 2, so it has to be the L0 value there.

Because it's gonna be two minus six, two minus four, two minus seven, so it's the first one.

And gallon one, So I don't know, the orangey one, 0 at 7, oh, it's not 0 at 6, so it has to be the one that corresponds to the 0.6. And you can see, for example, that this is one here. Alright, so the polynomial, for example, L0 is F2 is going to be 1, but you can see that before 2, so to the left, the value of the polynomial is larger than 1. So... One is not necessarily the highest value that the polynomial

of the Lagrange polynomial obtains. So did you guys ever work with Alagran school in Olmos before?

So they are well known. Because they're so nice. Okay. So In terms of construction, I can do it the hard way. Or the good way?

The hard way It's just as you see here. So the hard way, I mean the most costly one, I mean, in this case, it's not a big deal because it's a polynomial of four to three for four points, right? Even my calculator, I guess, could do it.

But the construction, the way you construct the polynomial, even if it's Lagrangian, you have to be smart about it.

So in this case it's direct construct.

So what I did here is I multiplied each one of the Lagrange polynomials by the value of the y's at the corresponding data points. So I have My data points were, in the Y's, were 14, 34, 35, and 50. So I'm going to multiply this first one by 14, the second one I'm going to multiply by 24, this one by 5, and the other three by 15, and this is what I have here. And so this is in terms of construction. And you can see that if you want to evaluate, if you don't do corners or anything, you will have n square floating point operations.

Every time you need to evaluate the value of the polynomial. And then the question is, can we do better? Yes, there's like a strategy to actually construct the Lagrange polynomials or a interpolating polynomial from Lagrange interpolation for Lagrange interpolators better. So the first thing that you do is you calculate The weights So you have, so the weights correspond, so they are constants. They only depend on the points that you are given. As you can see here, it was two, four, six.

six and seven. So the weights only depend on the points that you are given. And, um...

What you do is you calculate, of course, the weights just once.

And then you multiply the weights by each one of the y's at the particular point that you are considering, and you divide just by the term that it has the x in. Okay, so if you go here, If you go here, you don't have any x here in the denominator. You only have the x's on the numerator. But all of them have a certain pattern. So what we usually do is we construct a L that has all the terms, x minus 2, x minus all the points.

And then you just need to cancel the one term that corresponds not in the particularAnd if you do this, so this is the best construction. You can see the

construction is O of n squared. and which is, you know, much smaller Then the 2/3 of n cubed that we had before. with the 1x x squared and x cubed not so smart basis and the evaluation is over 500.

I just put the 5 so that you can do the count. I'm just counting computations. duplications, divisions, subtractions, and additions. So you can easilyOh, you're ahead. So this is like the strategy to come up with Laplace.

I have here in MATLAB implementation, because I just want you to be mind you of something that I usually forget.

I don't want you guys to do this. Is that if you go back to, so this is the MATLAB implementation.

have the polynomials, the Lagrangian evaluation. So you construct a polynomial in the equivalent and then you evaluate it. But I just want you guys to see one thing. What if, so this is the polynomial that I am talking about, what if I want to calculate the value of P3 at x equals to 2, for example? See, I ran into trouble here. And I want to remind you guys that if you're doing Lagrange interpolation, you have to account for the fact that eventually you can actually, you can possibly, let's say, need to calculate the value of the polynomial as at one point that you used to construct the point of view.

So in this case, if you go and calculate x, if you evaluate-So my T is my X, you see if you evaluate the T at XJ, you're gonna run into trouble because you have another number, right? So you usually, what you do is, you wanna make sure that this is an A, If it is exactly equal to x of j, usually you have a tolerance, but I just put it there so that you You can remind, that you can remember, that you have to make sure that if you run through all the points first, and if those points come, you have to make sure that you give the right value for the data without calculating, without evaluating the data.

Make sense? Yeah, so. The question is, this is my implementation of the Lagrange. And circulation. Which is this? So if I go for some reason I I go and I just want to calculate the value of p of 3, and x equals 2, I'm going to run into trouble, right? So what I usually do is a safeguard.

saying that if you are to evaluate the intercalant polynomial at any of the points that you use to construct it. So 2, 6, 4, and 7, instead of calculating using the formula, Lagrange's formula, you just say that it's equal to the value of y at that particular x. So this is what I've done here. So for all the J's, so for all the points that you have, If you are T, So if the value that you are evaluating the polynomial at is equal to the value of...

Is it OK? Yeah. I bet you don't need to interpolate. You don't use interpolation because-Doesn't make sense.

Okay, so let me kind of summarize. This Lagrange polynomial interpolation is not as simple as the regular monomials that we see. A is the identity. It's very stable for large N. And for Data that is very spread The construction cost is O of N squared, evaluation is O of N. Every time, this is O of n, every time you need to calculate, to evaluate the interpolant at a certain point, Now, the best part is that when you look at the interpulent, you know exactly what the coefficients multiplying the L's are.

Those are the values of the function that you are trying to evaluate at that particular point. The Again, I cannot emphasize this strongly enough Is that I constructed using the same data points?

Do it 30 times. Alright, using First using The wrong strategy Because these numbers here, they don't tell you anything about the data that you use to calculate the interpulental. So I calculated using this and then I calculated using different basis. So I've calculated using instead of 1 I use L0 instead of X I use L1 and so on and so forth.

The function, the polynomial that you get in the end is exactly the same because the interpolant is unique. Any questions regarding how this is, I mean, this is... This is really nice, I think. Okay, not so nice. So the idea is very simple, but not so nice is What errors?

Am I making or committing every time I use the interpulent instead of the function?

So what is the error? Questions that you might come up with is or are, what is the error in approximating a function by an interpolating polynomial.

You know, fair question.

How well so I know that At the data points, The F and the P have no error because it's an interpolating folding model.

But the question is, what is the error in between? Very fair question. If I have more notes Will I increase, if I have more data points, will I increase the quality of the interpolating polynomial? And then the last question, can I choose the location of those points, of the nodes, so that the error, Let's say that I'm making this smaller, which means that I'm increasing the quality of the interpolating point. So these are fair questions, I think.

I mean, I thought of all the questions that I could come up with, and maybe you have more, but I think those are-A very large or subset of all the questions that we, of all the reasonable questions that we might ask relating, interrelating polynomials. So there are two big theorems here. And there is a big question about whether or not the nodes are equally distanced from each other.

So those are two big things. We already know that they have to be distinct. Okay, so for example, If this is a question that you might come up with, I don't know how I came up with these numbers. I don't remember. But I would not have chosen these numbers right now.

What is the error? in replacing sine of x by its interpolating polynomial. And notice that I say by its interpolating polynomial because it's unique. Right? Constructed from 10 equally spaced nodes in that particular strange interval. So this is a question that I might be asking. Okay. And the first interpolating theorem Is this one. So I have As usual, x, I'm not, so I'm going to tell you that this theorem is Um...

The proof of this theorem is based on Rolle's theorem. I think it's section 10.3 or 10.4. of our free book. But this is not what I'm most interested in.

I'm mostly interested in that you understand what your theorem is saying and then using it afterwards. So there are n plus 1, again distinct nodes from x0 to xn. And they are all in a, b, including the a in the b. I have a function that is cn plus 1 in that interval a, b. And the P of n is the interpolating polynomial of degree less than or equal to n. of F at this particular point. Okay, so I have the function. I have, it has, I'm sorry, it has N plus one. Continue to the end.

In that interval, I have n plus 1 distinct nodes, and then I I have a formula That connects The function that I am interpolating at those points With the interpolating polynomial, so the formula says We have f of x. is equal to the interpolating polynomial plus a term. Okay, and this term, is such that If you give me it depends on the x, so the term is highly dependent on the x, Oh, not here. But it depends on the x through these terms here, but it also depends... On the X, because of the Kasai that you see here.

So the theorem says that for each x in a, b, there exists a number psi that depends on the x that you are considering. So the error is not the same for all the x's. In the interval. And this number psi is in between the nodes, so it's in between A and B, not including the A, not including B, such that this is true. Again, this is Mohr's theorem. And then you can say, oh, but this gives me a lot of information because if I want to remember, what do I want?

I want the error between the function and the interpolating polynomial, right? So if I want the error, what I do is I subtract From the left and from the right. F, so I subtract the polynomial, so F minus P of n. I take the absolute value, right? These are scalar functions and I have The errorYeah. I am committing and that error depends on x.

So what is the problem of this theorem? So it seems beautiful, right? It's like very beautiful. I can come up. This is what I was trying to say. I can come up here and just if I want the error, so this is my goal, I want the error here. I can just, oh, if I want the error, I go. and just calculate this would be The error? Right, so this is my error. Never.

Get. at x, so the error depends on the value. I know in particular that this error has to be zero at each one of the distinct points because the polynomial is interpolate, so it passes through the points.

So this is going to be equalDo this.

So f plus 1 psi of x. Well, I can put the n plus 1 out. Stereo. And then x minus x naught.

Dot dot dot x minus x. So what is the problem? What is missing from this theorem in some sense?

is that I don't know what the xi is. I don't know what that number is. So I I cannot calculate the error. I can calculate the bound on the error if I know the f. And I know, if I know the degree of the polynomial, this is the easy part.

This is the easy part, I can calculate this number.

If I know the function, I can differentiate it. But I don't know psi, which depends on the x, so I cannot calculate This number, but I can calculate the bomb if I know the answer. And of course if I know the distance between the X And the points, which I can because usually I know the x that I'm looking for and I know the points that I'm given. So this theorem is nice, but it's not very useful because I know nothing about the psi except that it stays in the interval and it depends on the x. But fear not.

If I had We have equally spaced nodes.

So if the nodes are equally spaced in the interval, So this means that to go From one note to the other, I just add age.

Okay, and I go from one door to the other. Then you can show... That the product of x minus xi is bounded by 1/4. H2N+1 N factorial. So this I know. It's not very obvious to know why this is true.

So I'm looking at... I am just looking at this part here. So this is my product.

From i equals to 0, you gotta use i's. M?

of X minus XI.

And they are telling me, so my result here is telling me that if I take the absolute value of this, it's going to be less than or equal to 1/4 And then...

H n plus 1 over multiplied by n factorial. I mean, it's more or less obvious There it should be, so if you have If this is A, if this is, so my A is X naught. If this is X, this, my A is X naught, is it? I don't remember. Yes.

My A is X naught, X1.

all the way to be here. So this would be x2.

If I have an x here, The distance between x and xi, so remember this here is h, So this distance is bounded by age over two.

Right, this is more or less reason.

So if you want it, You could go and say that Your productOkay.

This terms is going to be bounded by h n plus 1 over 2n plus 1. Right? But this is-So I would have this Instead of having this. So what's wrong with this calculation? with the reasoning that I was making here is that my X This is true for the fact that this here is true if your x is in between one, is only true if your x is in a certain interval. Because if you compare your x to x3, this This is larger than age over two. So this is not true for all... The intervals. Okay, so not true.

What is true is this. But again, this is only true if you have equally spaced intervals. So this takes about maybe 10, 15 minutes to show.

Okay, so if I... Go happily. into the first interpolating theorem.

So the first interpolation error theorem Doesn't require the nodes to be equally spaced but it requires the notes to be distinct. Now here comes the fun. Now I have I am working with equally spaced nodes. I mean, why would you do anything else?

I will tell you why you would do something else, but for now, let's assume, I mean, if you're calculating and interpolating a polynomial, and you can choose the points, why would you do anything different from equally spaced, right? So this is a reasonable way to go. It's not the best way as you will see, but it's a reasonable way to go.

Then I can go and update. Mike. The upper bound here, With this new fact that I just found, and I have, if I have a bound on the nth plus one derivative of the function at a given point, a bound m, then I can just upgrade and say that the error that I have is the same.

And this is Again, for equal this space, The second intercalation error theorem.

So if I go back, so this is like, The interpolating error, so the function, the difference between the function f And the interpolating polynomial is bounded by this value. So what did I do? Just going back.

I have this formula. I substituted this, I mean in absolute value, by 1 over 4 H n plus 1 n factorial.

I said this guy here in absolute value is also bounded by m. This N factorial cancels with all the terms of n plus 1 factorial except for the n plus 1. I have the 1/4 and I have the H.

And this is what I have here.

So if I go to If I go back, and then I'll stop here, to the example that I had, which was discuss the error If my sine of x is replaced by an interpolating polynomial constructed from 10 equally spaced nodes in that strange interval. So the function is sine. N plus 1 is the number of nodes that I have, which is 10. My a is 0 and my b is that number. Is that-Oh, that's the golden number, the 1 plus square root of 5 over 2.

I didn't recognize it dressed like this. Okay, now I see why I chose it. Now it makes sense. Okay, so if I take whatever derivative, in this case it's going to be n plus 1 equals to 10, it's a derivative of... So it's a cosine or a sine, or with a negative, but in absolute value.

A bound would be one. Right? So I can just plug in the numbers and get to this. to this error bot. So the error that you have in absolute value is less than or equal to 10 to the negative 9. Okay. So P9, remember, so if you have n plus 1 equals to 10, this means that your n is equal to 9. So the order of your interpolating polynomial is 9. So you call it P9. represents the sine, which is the function that you are trying to interpolate with an error of at most two times or two units in the ninth Decimal place.

which means that P9 approximates the sine with at least eight decimal digits of accuracy.

Which is pretty good. My thoughts.

So this is the type of Answers that you can give if you use this interpolating error Thank you. Okay, next time I will start, next time it will be on Wednesday, I will explain why we don't use equally spaced -